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POWDER CONTROL FOR ADDITIVE MANUFACTURING SYSTEMS

Abstract

An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component, a powder delivery device configured to direct a powder stream toward the melt pool, at least one sensor configured to generate sensor data representative of at least one process characteristic, and a computing device. The computing device is configured to receive the sensor data from the at least one sensor, determine, based on the sensor data and a predictive model data, at least one powder control parameter configured to achieve a predetermined powder feed rate of the powder stream, and control, based on the at least one powder control parameter, the energy delivery device and the powder delivery device to deposit a plurality of layers.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing techniques.

BACKGROUND

[0002] Additive manufacturing generates three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing may utilize powdered materials and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures. In addition to initial fabrication, additive manufacturing techniques may include subsequent machining processes to finish a component.

SUMMARY

[0003] The disclosure describes additive manufacturing systems, and methods for operating additive manufacturing systems, that monitor and control powder flow to influence additive process parameters. For example, powder feed rate affects process responses of an additive manufacturing process. At the same time, powder control parameters (e.g., powder size, powder density, carrier gas flow rate, or other parameters) and process characteristics (e.g., melt pool size) also affect powder flow. Thus, a predictive model may be used to predict, monitor, and/or control powder feed rate, accounting for other parameters and characteristics, to achieve a predetermined powder feed rate of the powder stream. In this way, the additive manufacturing systems described herein may enable more efficient use of powder in additive fabrication of a component.

[0004] In some examples, the disclosure describes an additive manufacturing system that includes an energy delivery device, a powder delivery device, at least one sensor, and a computing device. The energy delivery device is configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component. The powder delivery device is configured to direct a powder stream toward the melt pool. The at least one sensor is configured to generate sensor data representative of at least one process characteristic. The computing device is configured to receive the sensor data from the at least one sensor. The computing device is further configured to determine, based on the sensor data and a predictive model data, at least one powder control parameter configured to achieve a predetermined powder feed rate of the powder stream. The computing device is further configured to control, based on the at least one powder control parameter, the energy delivery device and the powder delivery device to deposit a plurality of layers.

[0005] In some examples, the disclosure describes a method that includes receiving, by a computing device, sensor data for a plurality of layers from at least one sensor of an additive manufacturing system, where the sensor data is representative of at least one process characteristic. The method may further include determining, by the computing device, based on the sensor data and a predictive model data, at least one powder control parameter configured to achieve a predetermined powder feed rate of the powder stream. The method may further include controlling, by the computing device, based on the at least one powder control parameter, an energy delivery device to deliver energy to a build surface to form a melt pool and a powder delivery device to direct a powder stream toward the melt pool.

[0006] The details of one or more examples are set forth in the accompanying drawings and the

description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 is a conceptual block diagram illustrating powder monitoring aspects of an example additive manufacturing system that includes a sensor configured to monitor at least one process characteristic of the additive manufacturing system during an additive manufacturing technique.

[0008] FIG. 2 is a conceptual and schematic diagram illustrating an example powder flow monitoring system configured to monitor powder flow between a powder delivery device and a build surface during the additive manufacturing technique.

[0009] FIG. 3 is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique.

[0010] FIG. 4 is a flowchart illustrating an example technique for fabricating a component while controlling a powder feed rate.

[0011] FIG. 5A is a conceptual diagram illustrating an example machine learning model.

[0012] FIG. 5B is a conceptual diagram illustrating an example training process for a machine learning model.

DETAILED DESCRIPTION

[0013] The disclosure generally describes techniques and systems for monitoring process characteristics indicative of powder feed rate during a blown powder additive manufacturing technique (e.g., directed energy deposition, DED), and controlling process characteristics based on a predictive model to achieve a predetermined powder feed rate. For example, the predetermined powder feed rate may promote capture of powder by a melt pool, and reduce underutilization of powder. Further, excess powder not captured by the melt pool may tend to collect in restricted, constrained, or narrow portions or regions of a component. Such collection of excess powder may increase machine wear and tear, loss of powder, and deviation from operational tolerances.

Controlling the powder feed rate to promote capture of a substantial portion of the powder by the melt pool, for example, substantially all powder by the melt pool (and eventually, into a fabricated component) may thus reduce material costs, processing costs, and machine maintenance and downtime.

[0014] During blown powder additive manufacturing, a component is built up by adding material to the component in sequential layers, such that the final component is composed of a plurality of layers of material. In blown powder additive manufacturing techniques for forming components from metals or alloys, an energy source may direct energy at a substrate to form a melt pool, and a powder delivery device may deliver a powder to the melt pool. At least some of the powder at least partially melts and is joined to the melt pool and, thus, the substrate.

[0015] Powder feed rate is a parameter that affects many process responses, such as build quality, build height, layer thickness, and additive processes may attempt to utilize a powder feed rate that promotes capture of substantially all powder directed to a melt pool by the melt pool, and ultimately, incorporation into a fabricated component. Excess powder may collect in constrained, narrow, or tight regions of components or substrates in additive manufacturing systems, which may affect efficiency, component life, product quality, and economical considerations.

[0016] Powder flow monitoring system and/or upstream powder monitoring such as gravimetric monitoring, and powder quality monitoring may be used to characterize the responsiveness of changing a powder feed rate, accounting for powder control parameters or variables (e.g., powder feeders, deposition heads, powder sizes, powder densities, machine designs, or carrier gas flow rates). A predictive model may be generated and used to predict, control, and or monitor powder

feed rate. For example, a modeled powder feed rate behavior may be used to determine mass flux based on part geometry (e.g., using deposit topology), melt pool capture capability (e.g., using melt pool size monitoring), and build strategy.

[0017] In accordance with techniques of this disclosure, an additive manufacturing system may include at least one sensor for measuring at least one process characteristic. By monitoring at least one process characteristic indicative of powder feed rate, the systems described herein may enable a more efficient use of powder in fabricating a component, more effectively and/or efficiently control operation of the machining device. For example, the at least one process characteristic and a predictive model may be used to control at least one powder control parameter configured to achieve a predetermined powder feed rate of the powder stream. The predetermined powder feed rate may promote utilization of powder into the component, and reduce loss or underutilization of powder.

[0018] FIG. 1 is a conceptual block diagram illustrating an example additive manufacturing system **10** configured to monitor and control a powder feed rate of a powder used to fabricate a component **22** during an additive manufacturing technique. In the example illustrated in FIG. 1, additive manufacturing system **10** includes a computing device **12**, a powder delivery device **14**, an energy delivery device **16**, a powder flow monitoring system (PFMS) **18**, a stage **20**, an optical system **50**, a melt pool monitoring system (MPMS) **52**, a powder source **42**, a powder source mass sensor **44**, and a sensor **48**.

[0019] Stage **20** is configured to position component **22** during an additive manufacturing process. In some examples, stage **20** is movable relative to energy delivery device **16** and/or energy delivery device **16** is movable relative to stage **20**. Similarly, stage **20** may be movable relative to powder delivery device **14** and/or powder delivery device **14** may be movable relative to stage **20**. Stage **20** may be configured to selectively position and restrain component **22** in place relative to stage **20** during manufacturing of component **22**.

[0020] Powder source **42** is the source of powder for powder stream **30**. Powder source **42** may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source **42** also may include a mechanism for entraining the powder in a gas flow. For instance, powder source **42** may be coupled to a gas source, which provides a gas flowing through powder source **42** and entraining powder within the gas flow. Additionally, or alternatively, powder source **42** may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream. System **10** may include a powder source mass sensor **44** associated with powder source **42**. Powder source mass sensor **44** may be configured to quantify loss of mass in the powder source **42** or, alternatively, a mass flow out of powder source **42**.

[0021] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. 1, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. 1, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0022] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder, such that the combined powder defines powder stream **30** focused at a focus plane. As powder delivery device **14** is movable in the z-axis shown in FIG. 1 relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with a build surface **28** of component **22**.

[0023] In some examples, powder delivery device **14** may be mechanically coupled or attached to

energy delivery device **16** to facilitate delivery of powder stream **30** and energy **34** for forming melt pool **32** to substantially the same location adjacent to component **22**. Energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy **34** that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0024] In some examples, energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**.

[0025] As shown in FIG. **1**, energy delivery device **16** may be arranged or configured such that energy **34** and powder stream **30** both exit from a common deposition head and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder nozzle(s) for forming powder stream **30** and directing powder stream **30** toward build surface **28**. At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**, and at least some of the powder that impacts melt pool **32** may be joined to component **22**.

[0026] During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Energy delivery device **16** may direct energy **34** at first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). The temperature and cooling rate of melt pool **32** and the surrounding areas of first layer **24** affect the microstructure of the component **22** formed using the additive manufacturing technique.

[0027] System **10** may include both mass flow monitoring and heat flow monitoring. To provide mass flow monitoring, system **10** may include powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light sources. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28**. The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such as greater than 1000 Hz. To provide heat flow monitoring, system **10** may include melt pool monitoring system (MPMS) **52**. MPMS **52** is configured to image at least a portion of melt pool **32** to detect parameters, such as size, temperature, or shape, of melt pool **32**. For example, MPMS may be communicatively coupled to optical system **50** for observing thermal emissions around melt pool **32** and a thermal camera for monitoring a size and/or temperature of melt pool **32**.

[0028] Optical system **50** may include an imaging device and an associated optical train, which

senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **50** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range. The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device.

[0029] System **10** further includes at least one sensor **48**. Sensor **48** may be configured to generate sensor data representative of at least one process characteristic of a process performed by system **10**. For example, the at least one process characteristic may include at least one of a powder flow rate, a carrier gas flow rate, a melt pool size, a powder quality, a powder velocity. Sensor **48** may include at least one of a powder flow monitor, a gas flow sensor, a melt pool monitor, a high speed camera, or a laser diffraction sensor.

[0030] Computing device **12** is configured to control components and operation of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** may be communicatively coupled to, and configured to control, powder delivery device **14**, energy delivery device **16**, PFMS **18**, optical system **50**, MPMS **52**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or sensor **48** using respective communication connections. Although FIG. **1** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**.

[0031] Computing device **12** may be configured to control operation of powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40**, stage **20**, and/or sensor **48** to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MPMS **52**, and/or sensor **48** during additive manufacturing processes. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40** and/or powder flow sensor to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MPMS **52**, and/or sensor **48**. Positioning component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MPMS **52**, and/or sensor **48** may include positioning a predetermined surface (e.g., a surface to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, MPMS **52**, and/or sensor **48**.

[0032] Computing device **12** may be configured to control system **10** to deposit layers **24** and **26** to form component **22** based on a set of deposition parameters. The set of deposition parameters may include energy, feed, and motion parameters that are configured to produce layers **24**, **26**, having various physical parameters, such as a height of layers **24**, **26**, and a density of layers **24**, **26**. For example, the set of deposition parameters may include power, beam diameter, beam profile, and wavelength of energy delivery device **16**; powder feed rate and gas feed rate of powder delivery device **14**; scan speed and deposition path of stage **20** relative to energy delivery device **16** and powder delivery device **14**; or any other operating parameters that may affect an amount and/or quality of material formed as layers **24**, **26**.

[0033] Computing device **12** may be configured to select deposition parameters that are configured to generate layers **24**, **26**, according to predetermined physical parameters. As shown in FIG. **1**, component **22** may include a first layer **24** and a second layer **26**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including first layer **24** and second layer **26**, the

technique may be extended to components **22** with more complex geometry and any number of layers.

[0034] To form component **22**, computing device **12** may control powder delivery device **14** and energy delivery device **16** according to the set of operating parameters to form, on a surface **28** of first layer **24** of material, a second layer **26** of material. Computing device **12** may control energy delivery device **16** to deliver energy **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**.

[0035] Computing device **12** may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** may then control a z-axis position of stage **20** and/or powder delivery device **14** and energy delivery device **16** such that melt pool **32** will be formed on surface **36** of second layer **26**, and may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete.

[0036] Computing device **12** may use sensor data from sensor **48** (alone, or in combination with sensor data from at least one other sensor) to control powder flow in system **10**, for example, to control a powder feed rate. For example, computing device **12** may be configured to receive sensor data from sensor **48** representative of at least one process characteristic. The at least one process characteristic may include any suitable process characteristic that may be associated with powder feed rate, including, for example, at least one of a powder flow rate, a carrier gas flow rate, a melt pool size, a powder quality, or a powder velocity. Computing device **12** may be further configured to determine, based on the sensor data and a predictive model data, at least one powder control parameter configured to achieve a predetermined powder feed rate of powder stream **30**. The predictive model data may be based on historical data or cohort data obtained from runs of system **10** or of other systems in which at least one powder control parameter influences a powder feed rate in system **10** having at least one process characteristic. For example, powder control parameters and may be varied, and the effect of changing one or more powder control parameters on powder feed rate may be measured and recorded for different magnitudes of process characteristics.

[0037] Computing device **12** (or some other computing device) may generate the predictive model based on powder control parameters, the process characteristics, and powder feed rates. For example, a machine learning model may be trained using any of powder control parameters, process characteristics, and powder feed rate as inputs or outputs, such that for a given combination of two of a powder control parameter, a process characteristic, and a powder feed rate, a third of the powder control parameter, a process characteristic, and a powder feed rate may be predicted. Thus, the trained machine learning model may be used as the predictive model. For example, computing device **12** may use sensor data from sensor **48** (indicative of at least one process characteristic) and a predetermined powder feed rate as inputs to the predictive model, and may determine at least one powder control parameter as an output of the predictive model. Accordingly, for a given process characteristic and a predetermined powder feed rate, computing device **12** may determine an appropriate value for the powder control parameter. Further, the predictive model may be used to determine either specific values, or ranges of values, for the powder control parameter, based on specific values, ranges or values, and tolerances, for at least one process characteristic and powder feed rate.

[0038] In some examples, computing device **12** itself generates the predictive model based on historical runs of system **10**. For example, computing device **12** may be further configured to

generate the predictive model data by determining a change in the powder feed rate or some other parameter associated with a change in the at least one powder control parameter. The other parameter may include at least one of carrier velocity, center purge gas velocity, particle velocity, powder stream profile, working powder spot size, working powder spot shape, working powder spot density profile, or nozzle type and geometry.

[0039] In other examples, the predictive model is generated by another computing device based on runs of other systems, or based on component modeling of system **10**.

[0040] The at least one powder control parameter may include, for example, at least one of a powder flow rate, a powder size, a powder density, a powder feeder geometry, a deposition head geometry, a carrier gas flow rate, a powder nozzle geometry, a center purge shielding gas flow rate, a nozzle standoff to substrate distance, a nozzle standoff to build surface distance, or any parameter that may affect the powder stream or powder working spot size, shape, or density profile.

[0041] Computing device **12** may be further configured to control, based on the at least one powder control parameter, energy delivery device **16** and powder delivery device **14**, to deposit a plurality of layers to form component **22**. Thus, computing device **12** may control or vary the at least one powder control parameter to ultimately achieve the predetermined powder feed rate. For example, the predetermined powder feed rate may promote the utilization of powder contacting melt pool **32** and being incorporated into component **22**, and thus reduce loss or underutilization of powder originating from powder source **42**.

[0042] Computing device **12** may be further configured to account for at least one component interaction parameter in determining a mass flux. For example, computing device **12** may be configured to determine, based on the at least one powder control parameter and at least one component interaction parameter, a mass flux; and control, based on the at least one powder control parameter and the mass flux, energy delivery device **16** and powder delivery device **14** to deposit the plurality of layers. The at least one component interaction parameter may be indicative of an interaction of one or both of powder or energy with melt pool **32**, or with component **22**, in course of additive manufacturing. For example, the at least one component interaction parameter may include at least one of a part geometry, a melt pool capture capability, or a tool path. Further, computing device **12** may be configured to determine at least one of the part geometry based on a deposit topology, the melt pool capture capability based on melt pool size, or the tool path based on a build strategy associated with component **22**.

[0043] Computing device **12** may also use sensor data from sensor **48** to determine a quality of component **22**, for example, conformance of or deviation of component **22** compared to geometric tolerances. For example, computing device **12** may be further configured to determine, based on the sensor data, at least one process response, and determine, by comparing the process response with a threshold response value, a quality index. The at least one process response may be indicative of an aspect of component **22** influenced by a process performed by system **10**. For example, the at least one process response may include at least one of a build quality, a build height, or a layer thickness.

[0044] FIG. **2** is a conceptual and schematic diagram illustrating an example powder flow monitoring system (PFMS) **60** configured to monitor powder flow between a powder delivery device **62** and a build surface (not shown in FIG. **2**) during an additive manufacturing technique. Powder delivery device **62** may be an example of powder delivery device **14** of FIG. **1**, and PFMS **60** may be an example of PFMS **18** of FIG. **1**.

[0045] Powder delivery device **62** includes a deposition head **64** that carries a plurality of powder nozzles **66**. Plurality of powder nozzles **66** output a powder stream **68** toward the build surface. As shown in FIG. **2**, powder stream **68** may be focused at a focal plane, such that powder stream **68** is converging toward the focal plane and diverging away from the focal plane.

[0046] PFMS **60** includes a housing **70** (also referred to as an enclosure), which encloses an imaging device **72** and an illumination device **74**. In some examples, imaging device **72** may be a

high-speed camera and illumination device **74** may be laser illuminator. Housing **70** is attached to an adjustable z-stage **76** by a bracket **78**.

[0047] Housing **70** is configured to enclose imaging device **72** and illumination device **74** and help protect imaging device **72** and illumination device **74** from a surrounding environment. For instance, housing **70** may be configured to surround imaging device **72** and illumination device **74** and prevent any powder that reflects from the build surface toward PFMS **60** from impacting imaging device **72** or illumination device **74**.

[0048] Further, housing **70** may be configured to cool imaging device **72** and illumination device **74**. Imaging device **72** and illumination device **74** may be exposed to heat from the melt pool at the build surface and energy from the energy delivery device. Imaging device **72** and illumination device **74** may be relatively sensitive to heat and have improved operational lifetime if maintained and operated below a certain temperature. PFMS **60** may include a cooling system **80** configured to remove heat from within housing **70** to cool imaging device **72** and illumination device **74**. For instance, cooling system **80** may include cooling fluid circuit through which a cooling fluid flows, and housing **70** may include part of the cooling circuit. In some examples, housing **70** may be formed from a material having relatively high thermal conductivity, such as aluminum, to help transfer heat from within housing **70** to cooling system **80** (e.g., a cooling fluid flowing through cooling system **80**).

[0049] As described above, PFMS **60** may be configured to measure powder flow of powder stream **68** (FIG. 2) at one or more axial (or longitudinal) locations of powder stream **68** and determine one or more parameters associated with the powder flow. For instance, illumination device **74** may illuminate powder of powder stream **68** in a plane oriented substantially orthogonal to a longitudinal axis that extends from powder delivery device **62** to the build surface. PFMS **60** may be positioned at a selected axial or longitudinal location to image a selected axial or longitudinal position between powder delivery device **62** and the build surface. Imaging device **72** may be configured to image at least some of the illuminated powder.

[0050] System **10** may be configured with various in-situ monitoring techniques, including powder flow monitoring, mass flux monitoring, and heat flux monitoring, to control an additive manufacturing process.

[0051] FIG. 3 is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique. The technique of FIG. 3 may be implemented by system **10** of FIG. 1 or system **60** of FIG. 2 and will be described with concurrent reference to FIGS. 1 and 2. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. 3 may be performed by other systems.

[0052] Computing device **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. 3). For instance, computing device **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**.

[0053] Computing device **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18** and/or powder source mass sensor **44**. Data received from powder source mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. In some examples computing device **12** may be configured to further receive powder flow data from sensor **48**. For example, sensor **48** may include a powder flow sensor, and data from sensor **48** may indicate a topology of build surface **28**, and may further indicate powder mass captured by melt pool **32** and added to component **22**.

[0054] Computing device **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18**, powder source mass sensor **44**, and/or sensor **48**. For example,

computing device **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32**, as determined based on data from powder flow sensor, into the mass flow determined based on data received from PFMS **18**. Further, computing device **12** may determine an overall mass flux using the data received from PFMS **18**, powder source mass sensor **44**, and/or sensor **48**. Computing device **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. **3**).

[0055] Similarly, computing device may be configured to control energy delivery device **16** to deliver energy **34** to first layer **24** to establish a given heat input (see bottom left of FIG. **3**). For example, one or more computing device **12** may control one or more operating parameters of energy delivery device **16**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**).

[0056] Computing device **12** may be configured to receive data from one or more heat sensors, such as optical system **50** and/or MPMS **52**. Computing device **12** may determine a cooling rate and associated heat from using data from optical system **50** and may determine a heat input into component using a size and/or temperature of melt pool **32** as observed by melt pool monitor. Computing device **12** may be configured to determine an overall heat flux using these data. Computing device **12** may then use the overall heat flux as an input to the control algorithm used to control the energy delivery by energy delivery device **16** (see bottom left of FIG. **3**). In some examples, computing device **12** may also use a deposit topology (captured powder mass) and/or capture efficiency metric in the determination of the heat flux, as the added powder mass and quench effects associated with the captured powder affect the cooling rate.

[0057] FIG. **4** is a flowchart illustrating an example technique for fabricating a component while controlling a powder feed rate. The technique of FIG. **4** may be implemented by system **10** of FIG. **1** or system **60** of FIG. **2** and will be described with concurrent reference to FIGS. **1** and **2**. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. **3** may be performed by other systems.

[0058] In some examples, the technique of FIG. **4** includes receiving, by computing device **12**, sensor data for a plurality of layers from at least one sensor **48** of additive manufacturing system **10** (**102**). The sensor data is representative of at least one process characteristic (associated with a process performed by system **10**). The at least one process characteristic may include at least one of a powder flow rate, a carrier gas flow rate, a melt pool size, a powder quality, or a powder velocity.

[0059] The technique may further include determining, by computing device **12**, based on the sensor data and a predictive model data, at least one powder control parameter configured to achieve a predetermined powder feed rate of powder stream **30** (**104**). The at least one powder control parameter may include at least one of a powder flow rate, a powder size, a powder density, a powder feeder geometry, a deposition head geometry, a carrier gas flow rate, a powder nozzle geometry, a center purge shielding gas flow rate, a nozzle standoff to substrate distance, a nozzle standoff to build surface distance, or any parameter that may affect the powder stream or powder working spot size, shape, or density profile.

[0060] The technique may further include controlling, by computing device **12**, based on the at least one powder control parameter, energy delivery device **16** to deliver energy **34** to build surface **28** to form melt pool **32** and powder delivery device **14** to direct powder stream **30** toward melt pool **32** (**106**).

[0061] The predictive model data may be generated by computing device **12** or any other computing device. In some examples, the technique includes, by computing device **12**, generating

the predictive model data by determining a change in the powder feed rate associated with a change in the at least one powder control parameter (**100**). Alternatively, computing device **12** may receive the predictive model data from another computing device, a server, a network, a memory storage device, or from a database.

[0062] The controlling (**106**) may include, or be augmented by, determining other parameters of system **10**. For example, the technique may further include determining, by computing device **12**, based on the at least one powder control parameter and at least one component interaction parameter, a mass flux, and controlling, based on the at least one powder control parameter and the mass flux, energy delivery device **16** and powder delivery device **14** to deposit the plurality of layers. The at least one component interaction parameter includes at least one of a part geometry, a melt pool capture capability, or a tool path. Further, the technique may include, by computing device **12**, at least one of: determining the part geometry based on a deposit topology; determining the melt pool capture capability based on melt pool size; or determining the tool path based on a build strategy.

[0063] Computing device **12** may further determine a quality index, for example, indicative of conformance of component **22** to predetermined tolerances. For example, the technique may further include, by computing device **12**, determining, based on the sensor data, at least one process response (**108**). Computing device **12** may further determine, by comparing the process response with a threshold response value, the quality index (**110**). The at least one process response may include at least one of a build quality, a build height, or a layer thickness.

[0064] As explained above, additive manufacturing systems described herein may be configured to use machine learning processes for evaluating sensor data to determine at least one powder control parameter and/or adjust the set of deposition parameters for subsequent layers. FIG. 5A is a conceptual diagram illustrating an example machine learning model according to one or more aspects of this disclosure. Machine learning model **200** may be an example of a machine learning model implemented by computing device **12**. Machine learning model **200** may be an example of a deep learning model, or deep learning algorithm, trained to determine deposition parameters for subsequent layers. Computing device **12** and/or another device, may train, store, and/or utilize machine learning model **200**, but other devices of system **10** may apply inputs to machine learning model **200**. In some examples, other types of machine learning and deep learning models or algorithms may be utilized. For example, a convolutional neural network model of ResNet-18 may be used. Some non-limiting examples of models that may be used for transfer learning include AlexNet, VGGNet, GoogleNet, ResNet50, or DenseNet, etc. Some non-limiting examples of machine learning techniques include Support Vector Machines, K-Nearest Neighbor algorithm, and Multi-layer Perceptron.

[0065] As shown in the example of FIG. 5A, machine learning model **200** may include three types of layers. These three types of layers include input layer **202**, hidden layers **204**, and output layer **206**. Output layer **206** includes the output from the transfer function **205** of output layer **206**. Input layer **202** represents each of the input values X1 through X4 provided to machine learning model **200**. In some examples, the input values may include any of the values input into the machine learning model, as described above. For example, the input values may include sensor data from sensor **48**, or other data received by computing device **12**, as described above. In addition, in some examples input values of machine learning model **200** may include additional data, such as other data that may be collected by or stored in system **200**.

[0066] Each of the input values for each node in the input layer **202** is provided to each node of a first layer of hidden layers **204**. In the example of FIG. 5A, hidden layers **204** include two layers, one layer having four nodes and the other layer having three nodes, but fewer or greater number of nodes may be used in other examples. Each input from input layer **202** is multiplied by a weight and then summed at each node of hidden layers **204**. During training of machine learning model **200**, the weights for each input are adjusted to establish a relationship between topological data and

deposition parameters that generate the layer represented by the topological data. In some examples, one hidden layer may be incorporated into machine learning model **200**, or three or more hidden layers may be incorporated into machine learning model **200**, where each layer includes the same or different number of nodes.

[0067] The result of each node within hidden layers **204** is applied to the transfer function of output layer **206**. The transfer function may be linear or non-linear, depending on the number of layers within machine learning model **200**. Example non-linear transfer functions may be a sigmoid function or a rectifier function. The output **207** of the transfer function may be a set of deposition parameters that correspond to the layer represented by the topological data and having a desired thickness.

[0068] As shown in the example above, by applying machine learning model **200** to input data such as image data, processing circuitry of computing device **12** is able to determine a relationship between deposition parameters that produce a layer based on the sensor data. Computing device **12** may utilize such features to generate a predictive model and/or use the predictive model to determine at least one powder control parameter based on sensor data.

[0069] FIG. **5B** is a conceptual diagram illustrating an example training process for a machine learning model according to one or more aspects of this disclosure. Process **210** may be used to train machine learning model **200**. Machine learning model **200** may be implemented using any number of models for supervised and/or reinforcement learning, such as but not limited to, an artificial neural network, a decision tree, naïve Bayes network, support vector machine, or k-nearest neighbor model, CNN, RNN, LSTM, ensemble network, to name only a few examples.

[0070] In some examples, computing device **12** or another device trains machine learning model **100** based on a corpus of training data **212**. Training data **212** may include, for example, previous topological data of a build over time, previous deposition parameter data, and/or the like. Previous deposition parameter data, for example, may include deposition parameters associated with previous additive manufacturing processes. In some examples, the sensor data and/or deposition parameter data may be generated by executing a natural language processing application on content of a plurality of operational records to automatically extract relevant nor desired training data. For example, by using an NLP model, computing device **12** may capture potentially confounding operation conditions or factors, which may be useful in determining a set of deposition parameters. Computing device **12** may use the NLP model to capture such details.

[0071] In some examples, training data **212** may include annotations identifying effects of deposition parameters indicated in topological data of training data **212**. Training data **212** may include data from past additive deposition processes performed on components having different geometries, formed from different materials, formed under different conditions, and/or the like.

[0072] While training machine learning model **200**, computing device **12** may compare **214** a prediction or classification with a target output **216**. Computing device **12** may utilize an error signal from the comparison to train (learning/training **218**) machine learning model **200**. Computing device **12** may generate machine learning model weights or other modifications which computing device **12** may use to modify machine learning model **200**. For example, computing device **12** may modify the weights of machine learning model **200** based on the learning/training **218**.

[0073] The following enumerated clauses describe various examples according to the present disclosure.

[0074] Clause 1: An additive manufacturing system, including an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; at least one sensor configured to generate sensor data representative of at least one process characteristic; and a computing device configured to: receive the sensor data from the at least one sensor; determine, based on the sensor data and a predictive model data, at least one

powder control parameter configured to achieve a predetermined powder feed rate of the powder stream; and control, based on the at least one powder control parameter, the energy delivery device and the powder delivery device to deposit a plurality of layers.

[0075] Clause 2: The additive manufacturing system of clause 1, where the at least one sensor includes at least one of a powder flow monitor, a gas flow sensor, a melt pool monitor, a high speed camera, or a laser diffraction sensor.

[0076] Clause 3: The additive manufacturing system of clauses 1 or 2, where the at least one process characteristic includes at least one of a powder flow rate, a carrier gas flow rate, a melt pool size, a powder quality, or a powder velocity.

[0077] Clause 4: The additive manufacturing system of any of clauses 1 to 3, where the at least one powder control parameter includes at least one of a powder flow rate, a powder size, a powder density, a powder feeder geometry, a deposition head geometry, a carrier gas flow rate, a powder nozzle geometry, a center purge shielding gas flow rate, a nozzle standoff to substrate distance, a nozzle standoff to build surface distance, or any parameter that may affect the powder stream or powder working spot size, shape, or density profile.

[0078] Clause 5: The additive manufacturing system of clause 4, where the computing device is further configured to generate the predictive model data by determining a change in the powder feed rate associated with a change in the at least one powder control parameter.

[0079] Clause 6: The additive manufacturing system of any of clauses 1 to 5, where the computing device is further configured to: determine, based on the at least one powder control parameter and at least one component interaction parameter, a mass flux; and control, based on the at least one powder control parameter and the mass flux, the energy delivery device and the powder delivery device to deposit the plurality of layers.

[0080] Clause 7: The additive manufacturing system of clause 6, where the at least one component interaction parameter includes at least one of a part geometry, a melt pool capture capability, or a tool path.

[0081] Clause 8: The additive manufacturing system of clause 7, where the computing device is configured to determine at least one of the part geometry based on a deposit topology, the melt pool capture capability based on melt pool size, or the tool path based on a build strategy.

[0082] Clause 9: The additive manufacturing system of any of clauses 1 to 8, where the computing device is further configured to: determine, based on the sensor data, at least one process response; and determine, by comparing the process response with a threshold response value, a quality index.

[0083] Clause 10: The additive manufacturing system of clause 9, where the at least one process response includes at least one of a build quality, a build height, or a layer thickness.

[0084] Clause 11: A method for additive manufacturing, including: receiving, by a computing device, sensor data for a plurality of layers from at least one sensor of an additive manufacturing system, where the sensor data is representative of at least one process characteristic; determining, by the computing device, based on the sensor data and a predictive model data, at least one powder control parameter configured to achieve a predetermined powder feed rate of the powder stream; and controlling, by the computing device, based on the at least one powder control parameter, an energy delivery device to deliver energy to a build surface to form a melt pool and a powder delivery device to direct a powder stream toward the melt pool.

[0085] Clause 12: The method of clause 11, where the at least one sensor includes at least one of a powder flow monitor, a gas flow sensor, a melt pool monitor, a high speed camera, or a laser diffraction sensor.

[0086] Clause 13: The method of clauses 11 or 12, where the at least one process characteristic includes at least one of a powder flow rate, a carrier gas flow rate, a melt pool size, a powder quality, or a powder velocity.

[0087] Clause 14: The method of clause 13, where the at least one powder control parameter includes at least one of a powder flow rate, a powder size, a powder density, a powder feeder

geometry, a deposition head geometry, a carrier gas flow rate, a powder nozzle geometry, a center purge shielding gas flow rate, a nozzle standoff to substrate distance, a nozzle standoff to build surface distance, or any parameter that may affect the powder stream or powder working spot size, shape, or density profile.

[0088] Clause 15: The method of any of clauses 11 to 14, further including, by the computing device, generating the predictive model data by determining a change in the powder feed rate associated with a change in the at least one powder control parameter.

[0089] Clause 16: The method of any of clauses 11 to 15, further including, by the computing device: determining, based on the at least one powder control parameter and at least one component interaction parameter, a mass flux; and controlling, based on the at least one powder control parameter and the mass flux, the energy delivery device and the powder delivery device to deposit the plurality of layers.

[0090] Clause 17: The method of clause 16, where the at least one component interaction parameter includes at least one of a part geometry, a melt pool capture capability, or a tool path.

[0091] Clause 18: The method of clause 17, further including, by the computing device,

[0092] at least one of: determining the part geometry based on a deposit topology; determining the melt pool capture capability based on melt pool size; or determining the tool path based on a build strategy.

[0093] Clause 19: The method of any of clauses 11 to 18, further including, by the computing device: determining, based on the sensor data, at least one process response; and determining, by comparing the process response with a threshold response value, a quality index.

[0094] Clause 20: The method of clause 19, where the at least one process response includes at least one of a build quality, a build height, or a layer thickness.

[0095] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0096] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0097] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions.

Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors.

Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a

hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0098] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0099] Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. An additive manufacturing system, comprising: an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; at least one sensor configured to generate sensor data representative of at least one process characteristic; and a computing device configured to: receive the sensor data from the at least one sensor; determine, based on the sensor data and a predictive model data, at least one powder control parameter configured to achieve a predetermined powder feed rate of the powder stream; and control, based on the at least one powder control parameter, the energy delivery device and the powder delivery device to deposit a plurality of layers.
2. The additive manufacturing system of claim 1, wherein the at least one sensor comprises at least one of a powder flow monitor, a gas flow sensor, a melt pool monitor, a high speed camera, or a laser diffraction sensor.
3. The additive manufacturing system of claim 1, wherein the at least one process characteristic comprises at least one of a powder flow rate, a carrier gas flow rate, a melt pool size, a powder quality, or a powder velocity.
4. The additive manufacturing system of claim 1, wherein the at least one powder control parameter comprises at least one of a powder flow rate, a powder size, a powder density, a powder feeder geometry, a deposition head geometry, a carrier gas flow rate, a powder nozzle geometry, a center purge shielding gas flow rate, a nozzle standoff to substrate distance, a nozzle standoff to build surface distance, or any parameter that may affect the powder stream or powder working spot size, shape, or density profile.
5. The additive manufacturing system of claim 4, wherein the computing device is further configured to generate the predictive model data by determining a change in the powder feed rate associated with a change in the at least one powder control parameter.
6. The additive manufacturing system of claim 1, wherein the computing device is further configured to: determine, based on the at least one powder control parameter and at least one component interaction parameter, a mass flux; and control, based on the at least one powder control parameter and the mass flux, the energy delivery device and the powder delivery device to deposit the plurality of layers.
7. The additive manufacturing system of claim 6, wherein the at least one component interaction parameter comprises at least one of a part geometry, a melt pool capture capability, or a tool path.
8. The additive manufacturing system of claim 7, wherein the computing device is configured to determine at least one of the part geometry based on a deposit topology, the melt pool capture capability based on melt pool size, or the tool path based on a build strategy.
9. The additive manufacturing system of claim 1, wherein the computing device is further configured to: determine, based on the sensor data, at least one process response; and determine, by comparing the process response with a threshold response value, a quality index.
10. The additive manufacturing system of claim 9, wherein the at least one process response

comprises at least one of a build quality, a build height, or a layer thickness.

11. A method for additive manufacturing, comprising: receiving, by a computing device, sensor data for a plurality of layers from at least one sensor of an additive manufacturing system, wherein the sensor data is representative of at least one process characteristic; determining, by the computing device, based on the sensor data and a predictive model data, at least one powder control parameter configured to achieve a predetermined powder feed rate of the powder stream; and controlling, by the computing device, based on the at least one powder control parameter, an energy delivery device to deliver energy to a build surface to form a melt pool and a powder delivery device to direct a powder stream toward the melt pool.

12. The method of claim 11, wherein the at least one sensor comprises at least one of a powder flow monitor, a gas flow sensor, a melt pool monitor, a high speed camera, or a laser diffraction sensor.

13. The method of claim 11, wherein the at least one process characteristic comprises at least one of a powder flow rate, a carrier gas flow rate, a melt pool size, a powder quality, or a powder velocity.

14. The method of claim 13, wherein the at least one powder control parameter comprises at least one of a powder flow rate, a powder size, a powder density, a powder feeder geometry, a deposition head geometry, a carrier gas flow rate, a powder nozzle geometry, a center purge shielding gas flow rate, a nozzle standoff to substrate distance, a nozzle standoff to build surface distance, or any parameter that may affect the powder stream or powder working spot size, shape, or density profile.

15. The method of claim 11, further comprising, by the computing device, generating the predictive model data by determining a change in the powder feed rate associated with a change in the at least one powder control parameter.

16. The method of claim 11, further comprising, by the computing device: determining, based on the at least one powder control parameter and at least one component interaction parameter, a mass flux; and controlling, based on the at least one powder control parameter and the mass flux, the energy delivery device and the powder delivery device to deposit the plurality of layers.

17. The method of claim 16, wherein the at least one component interaction parameter comprises at least one of a part geometry, a melt pool capture capability, or a tool path.

18. The method of claim 17, further comprising, by the computing device, at least one of: determining the part geometry based on a deposit topology; determining the melt pool capture capability based on melt pool size; or determining the tool path based on a build strategy.

19. The method of claim 11, further comprising, by the computing device: determining, based on the sensor data, at least one process response; and determining, by comparing the process response with a threshold response value, a quality index.

20. The method of claim 19, wherein the at least one process response comprises at least one of a build quality, a build height, or a layer thickness.
